Daniel Mark Pantaleo

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EDUCATION

RMIT University **Graduated December 2025**

Bachelor of Engineering (Honours) – Advanced Manufacturing and Mechatronics

- Honours Class 2A, WAM 78, GPA 3.3 (4.0 grade scale).

SKILLS

CAD: SolidWorks (CSWA), Bambu Studio.

Programming/IT: Python, C, C++, MATLAB, Simulink, Maplesoft, Computer Vision, PCBs, PLCs, MS Office Suite, Teams.

Fabrication: 3D-printing, Board Prototyping, Soldering, Toolpath Planning.

PROJECTS

Capstone Project **Completed 2025**

- Team of six, tasked with developing an imaging system for a novel rheometer prototype to validate the difference in position of two oscillating pistons and the silicone seals attached to them using object identification and motion tracking algorithms.
- The industry partner required a faster, more accurate alternative to their manual image analysis methods.
- Designed and built a rig consisting of a 3D-printed base to stabilise and house the microscope cameras, rheometer prototype and required circuitry.
- Programmed an intuitive GUI that uses synchronised recording and lens calibration algorithms to collect a data set, then uses template matching and sparse optical flow tracking methods to accurately track the piston and seal positions. The obtained data is plotted, saved locally and displayed to the user in the program.
- Evaluated tracking accuracy as 98.9% using Pearson Correlation when the object tracking results from our solution was matched with the results from a linear encoder.
- Removal of manual work resulted in a reduction in overall analysis time by 44%.
- Received top project award for mechatronics at RMIT's annual engineering project expo, EnGenius.

Course Project **Completed 2024**

- The project required designing and 3D-printing a gripper arm mechanism and programming an integrated circuit to automatically detect and grip an egg within a certain force range.
- Selected a lead screw design to create a very high torque and allow for precise movement of the end effector.
- Through iterative design, specific volumes of the gripper arms were printed with higher density to prevent fractures caused by excessive stress. The final design could be cycled reliably without any signs of failure.
- The final design used PID control with a DC motor, load cell and IR sensor for position detection and precise force control, capable of creating a force within the project's requirements.

AWARDS / CERTIFICATES

Airmaster Mechatronics Emerging Innovators Award **Received 2025**

- Awarded to the top project in Manufacturing, Mechatronics and Materials Engineering at EnGenius 2025, sponsored by Airmaster.

Certified SolidWorks Associate (CSWA) **Received 2026**

- Obtained while teaching myself SolidWorks. I plan to obtain the professional certificate in future.

WORK EXPERIENCE

BWS **2023 - 2026**

Sales Assistant – Part Time

- Demonstrated strong interpersonal skills by engaging with customers and providing great service.
- Utilised organisational and detail-oriented skills while managing inventory and handling money.